

Experience on Seismic Retrofitting of Buildings in Nepal



NSET

HRRP-Technological Session on Retrofitting

22nd November 2016

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Overview of Presentation

- **Analysis Methods in Retrofitting**
- **Retrofitting methods of masonry buildings**
- **Retrofitting methods of RC buildings**



Introduction

(Source: Earthquake Protection)

Various study reports reveal more than 60% of the buildings in Kathmandu Valley are unsafe at MMI IX intensity earthquake

NSET experience in assessing thousands of buildings indicate most of the masonry Buildings and 40% of RC buildings (NSET assessed buildings only) are unsafe

Gorkha earthquake shows many buildings damaged even at MMI VI and MMI VII Intensity earthquake

**NO OTHER OPTION THAN STRENGTHENING EXISTING
VULNERABLE BUILDINGS**

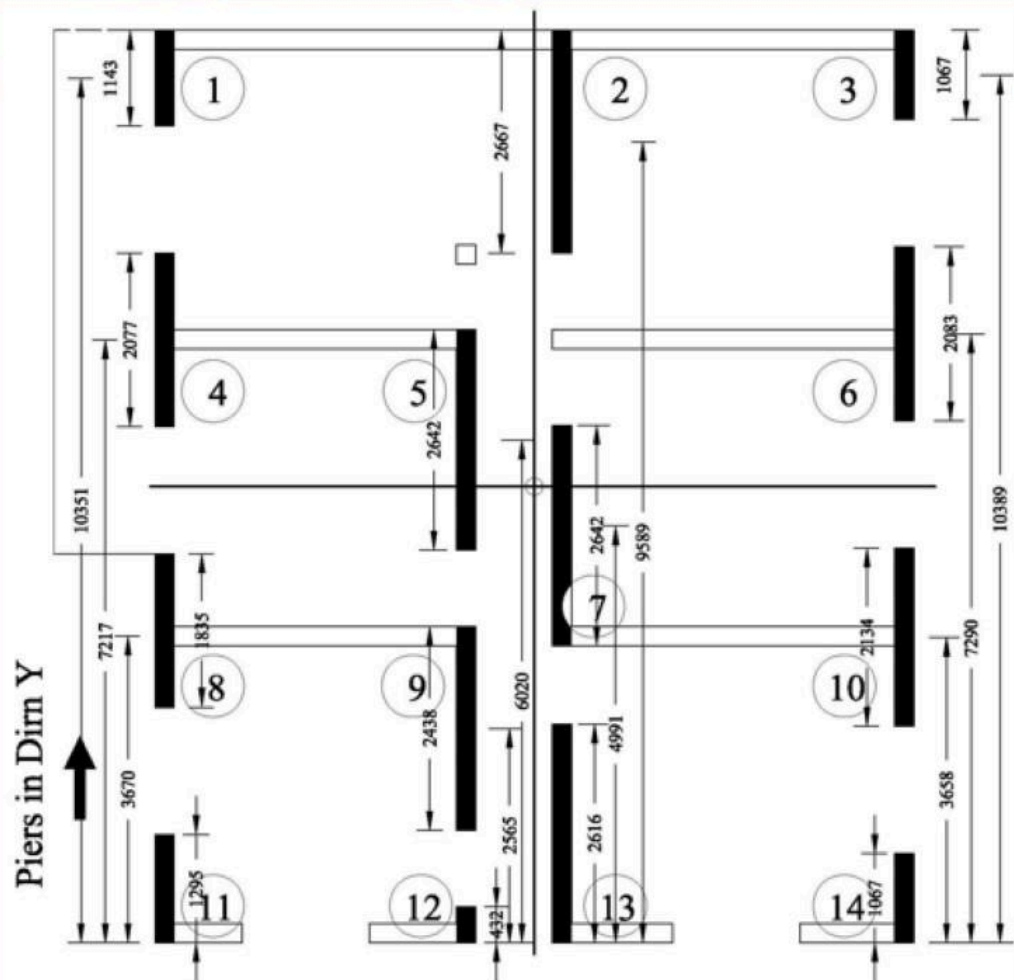


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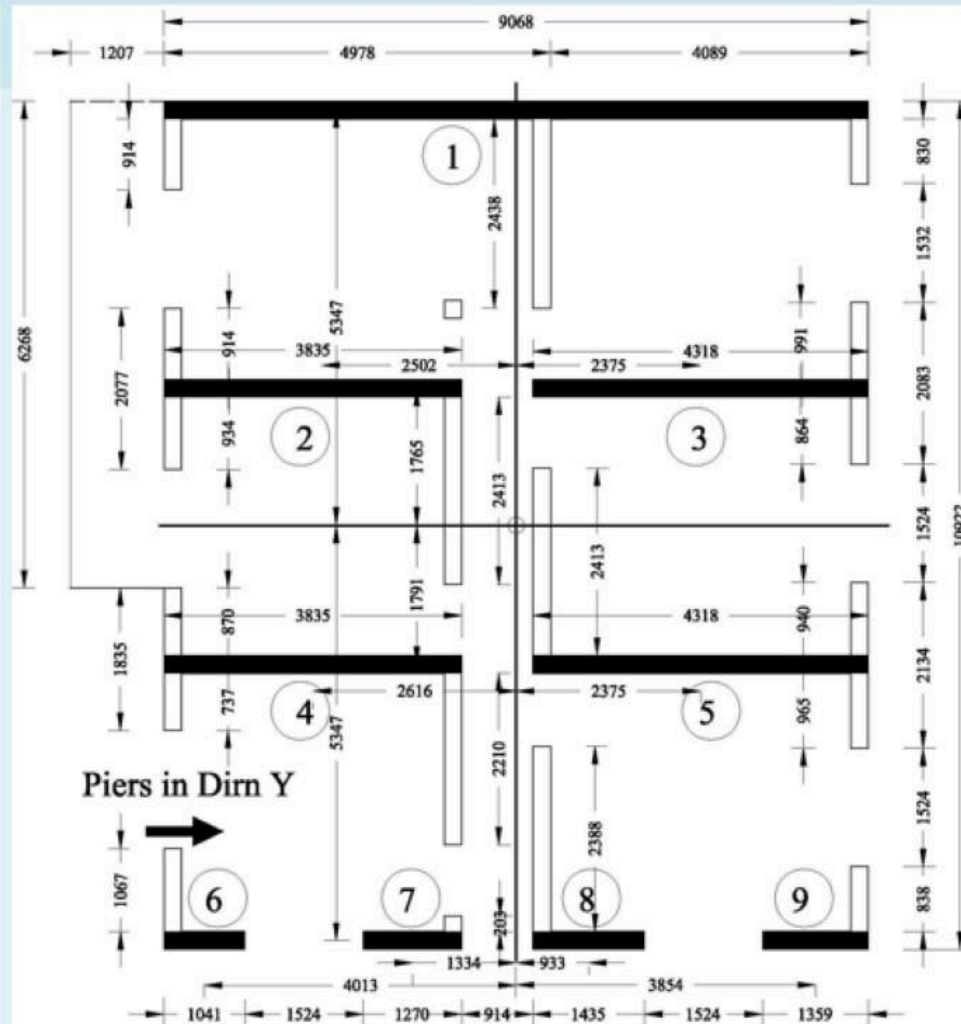
Analysis and Design of Retrofitting

- **Analysis Methods**
- **Retrofitting Options**

Analysis Methods: Masonry



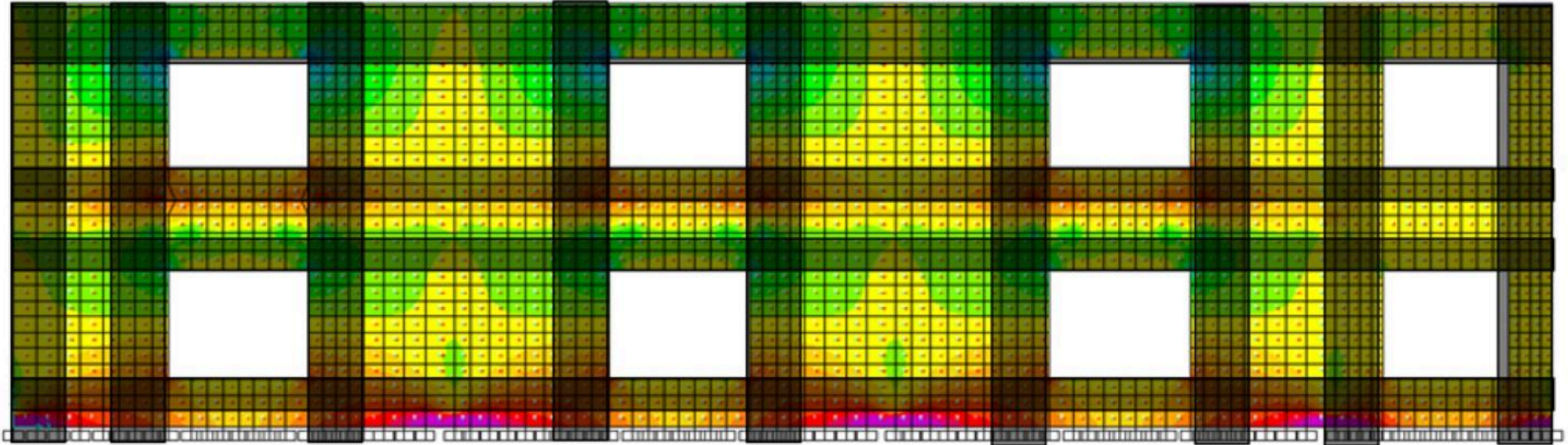
Ground Floor Plan



Ground Floor Plan

1. Individual Piers Analysis

Masonry: Analysis Methods

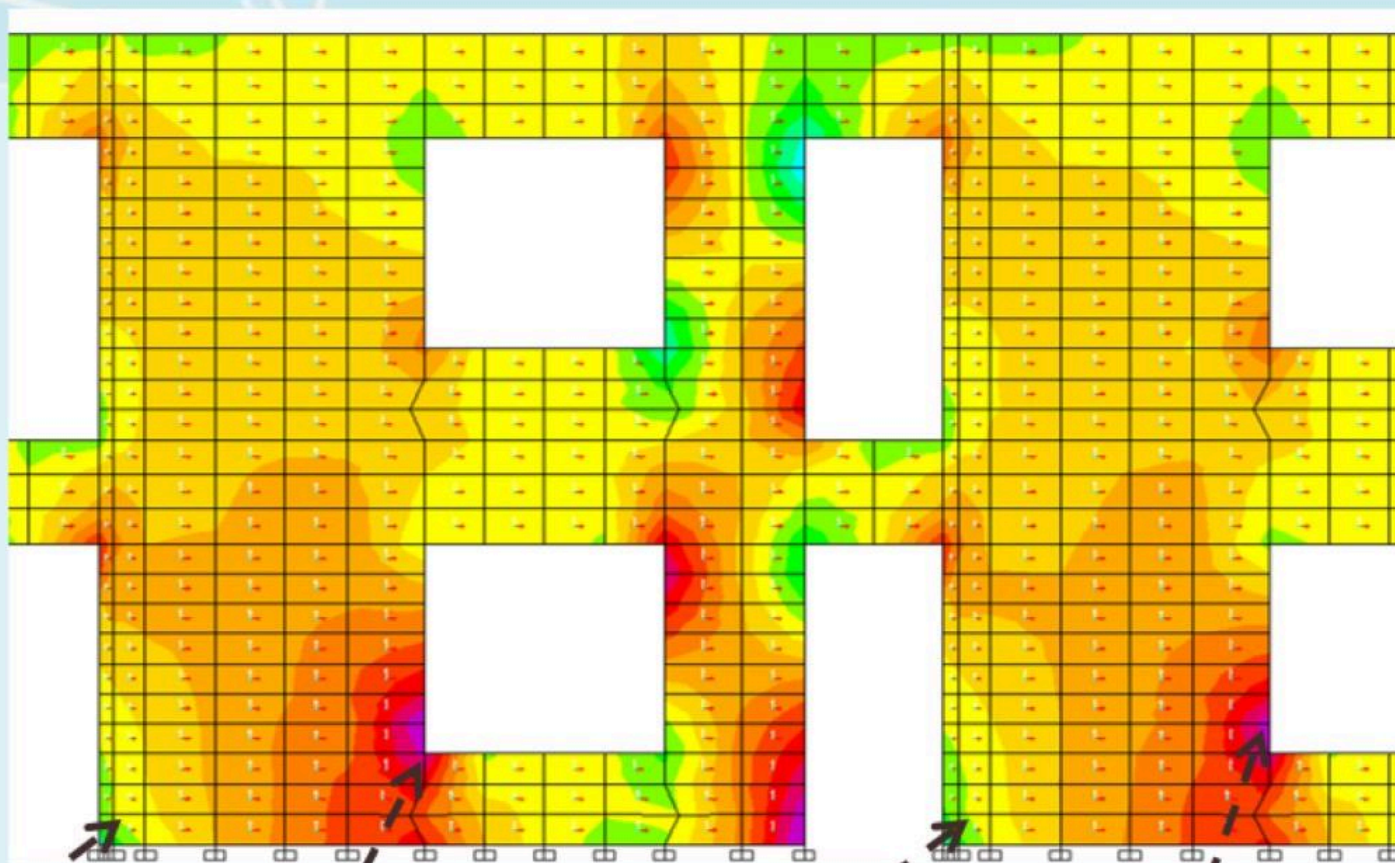


2. Elastic Analysis



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Masonry: Analysis Compression and Tensile Stress Distribution in Wall Panel



Tension
Zone

Compression
Zone

Tension
Zone

Compression
Zone

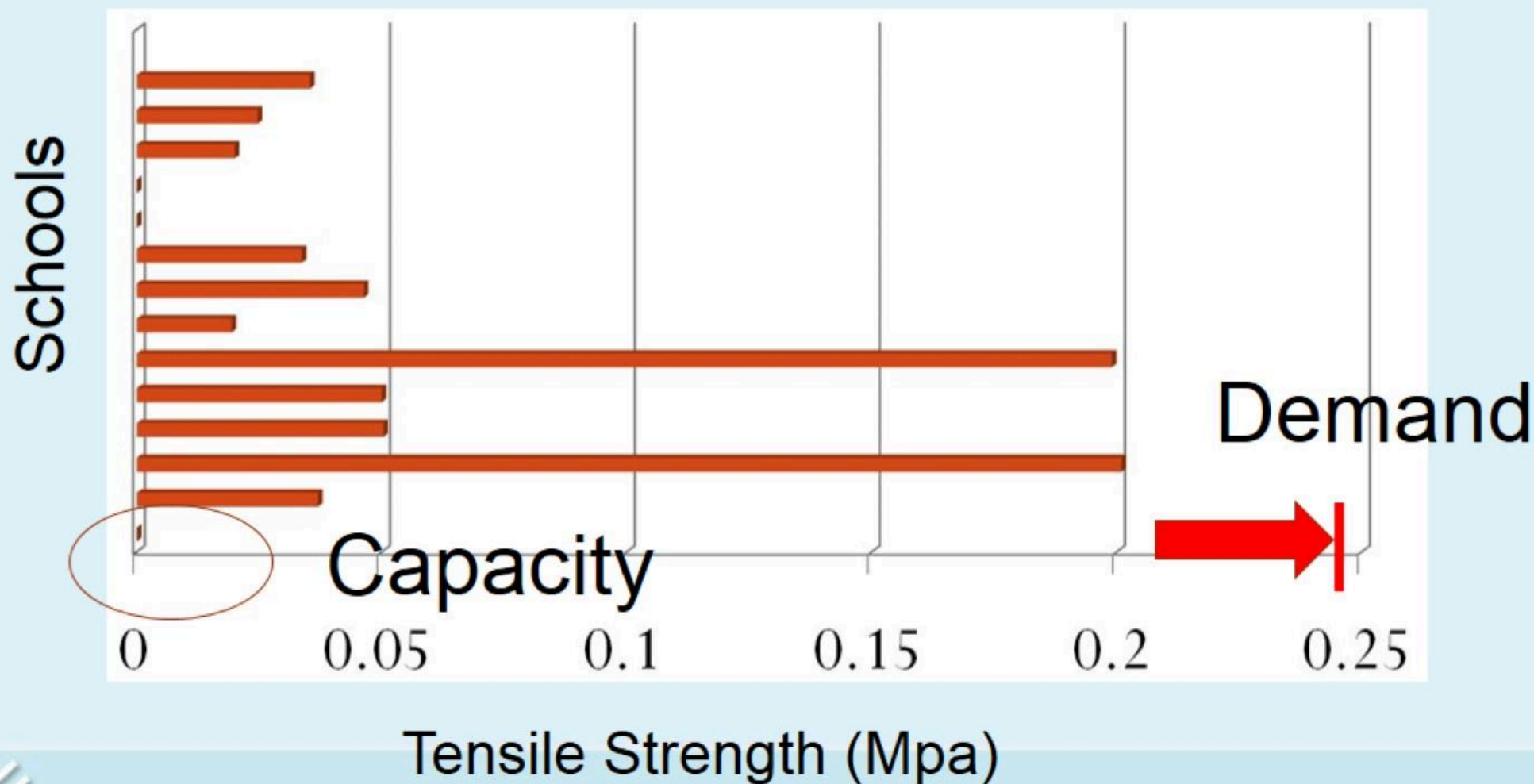
Maximum tensile
stress induced
(Adarsha HSS)=0.24
 N/mm^2

Existing tensile
strength is negligible

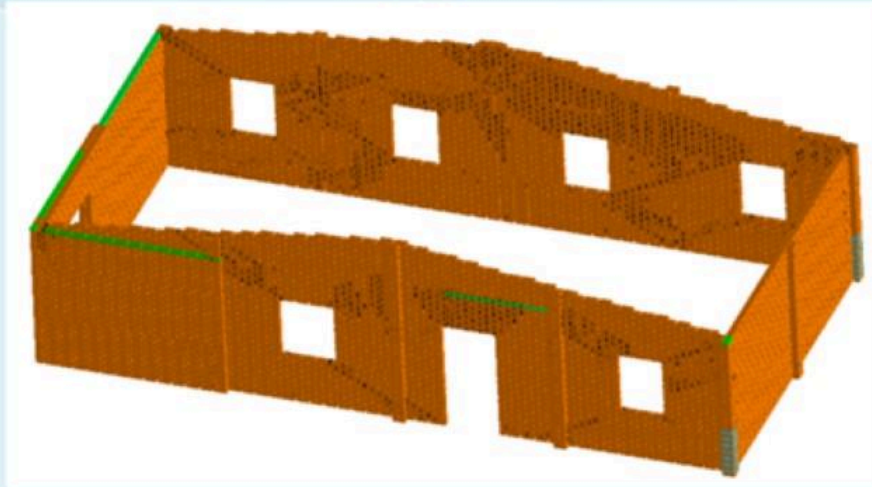


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Demand Vs Existing Capacity

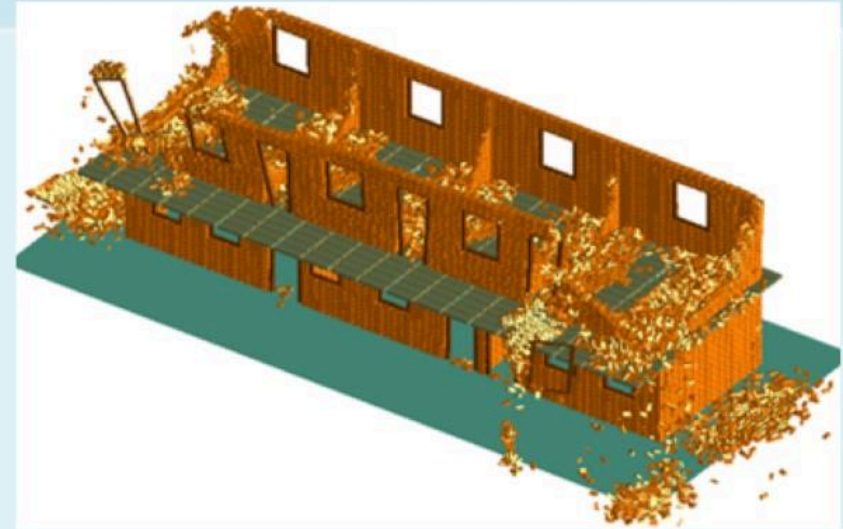


Masonry: Analysis Methods



Moderate Damage

- 1 story
- Average cement mortar
- 0.2g PGA



Complete Damage

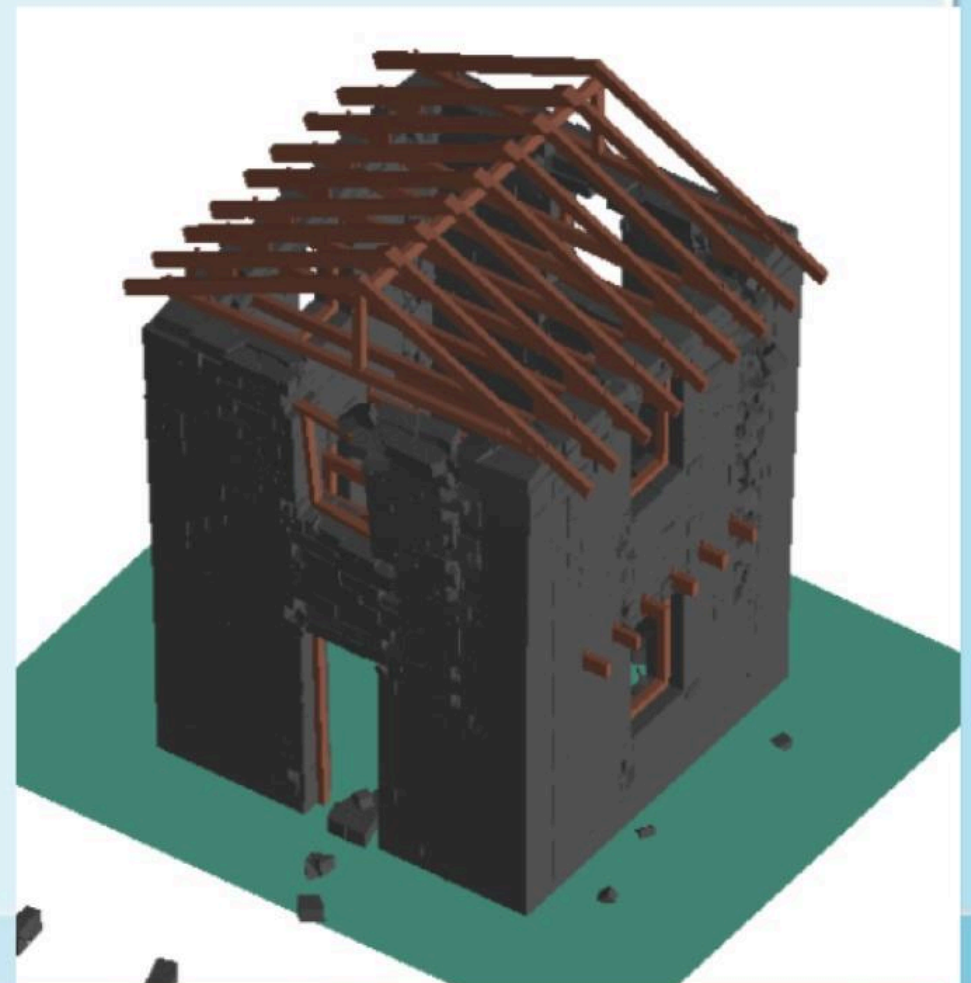
- 2 story
- Mud weak mortar
- Flexible roof
- 0.2g PGA

3. Non-linear Analysis and Verification



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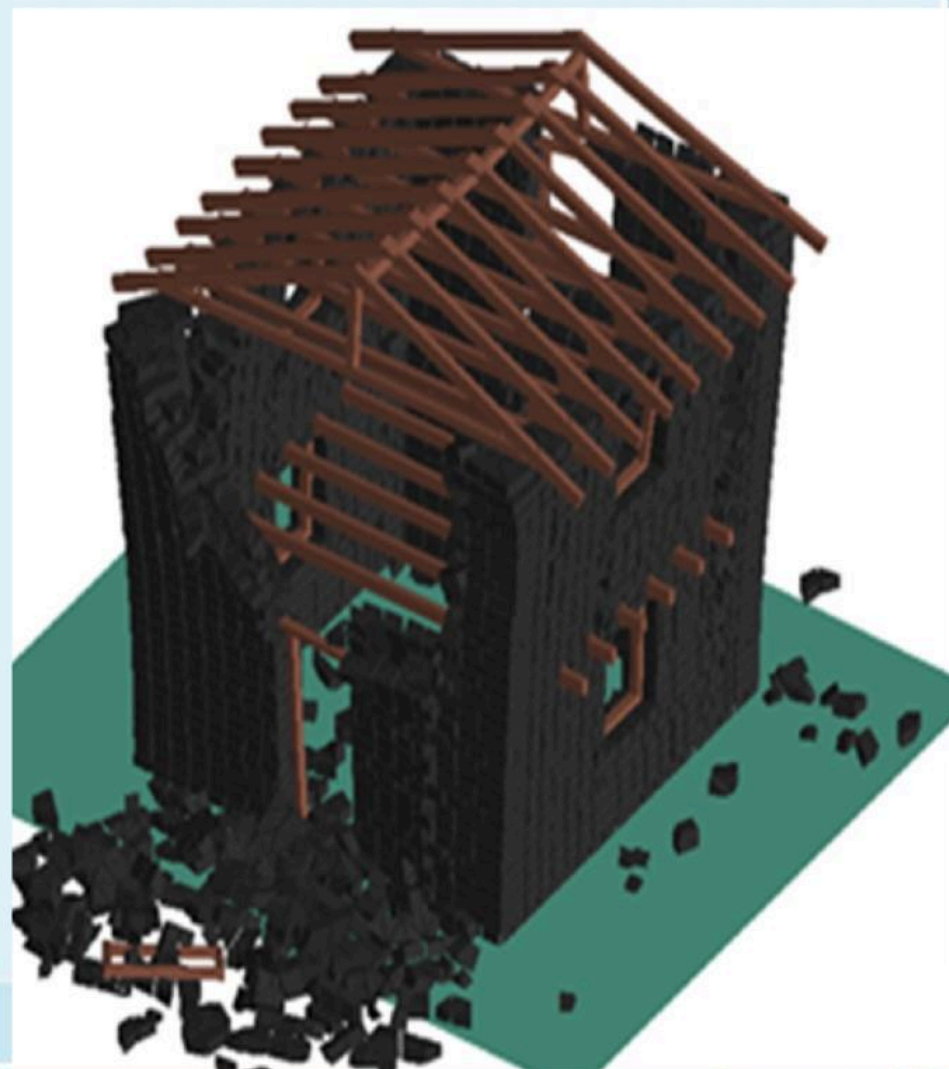
Laboratory Tests and Analytical Verification 0.3g for Stone Masonry



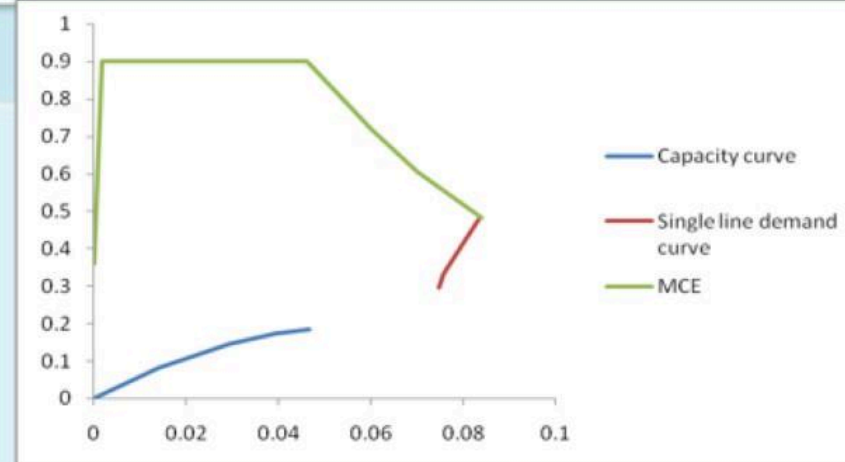
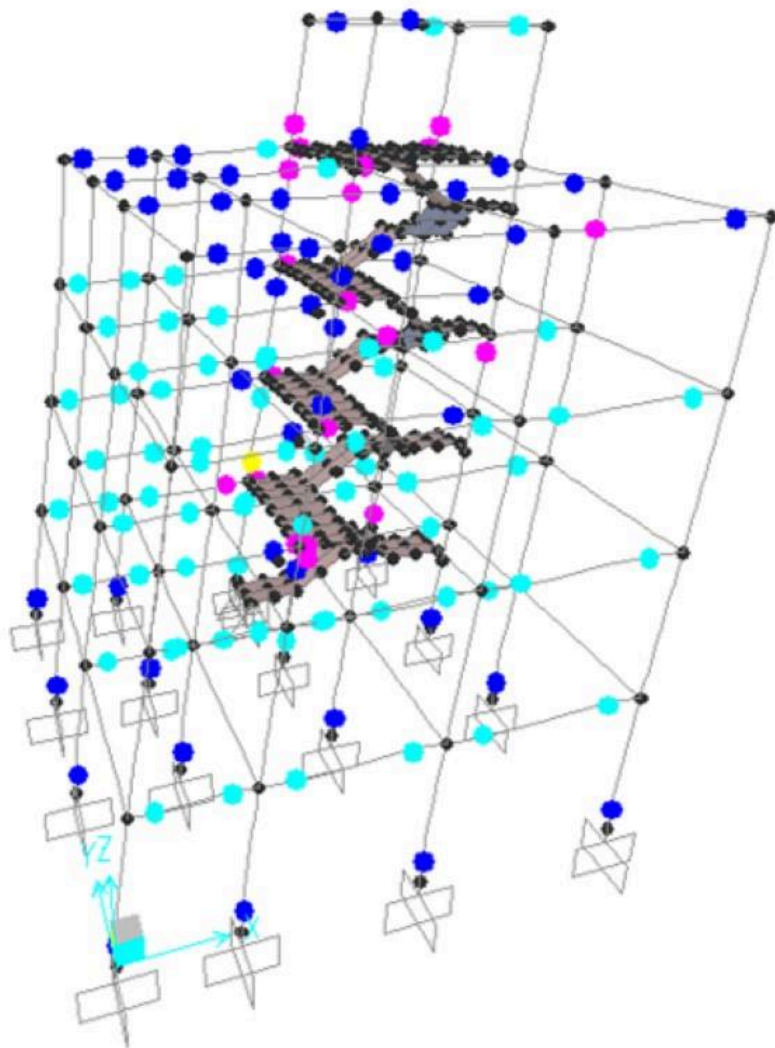


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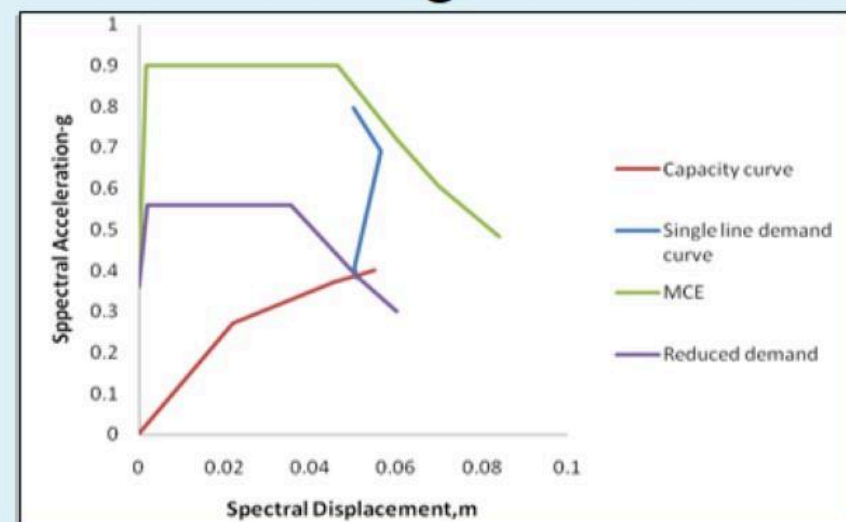
Shake Table Test Result: 0.4g



RC Buildings: Pushover Analysis

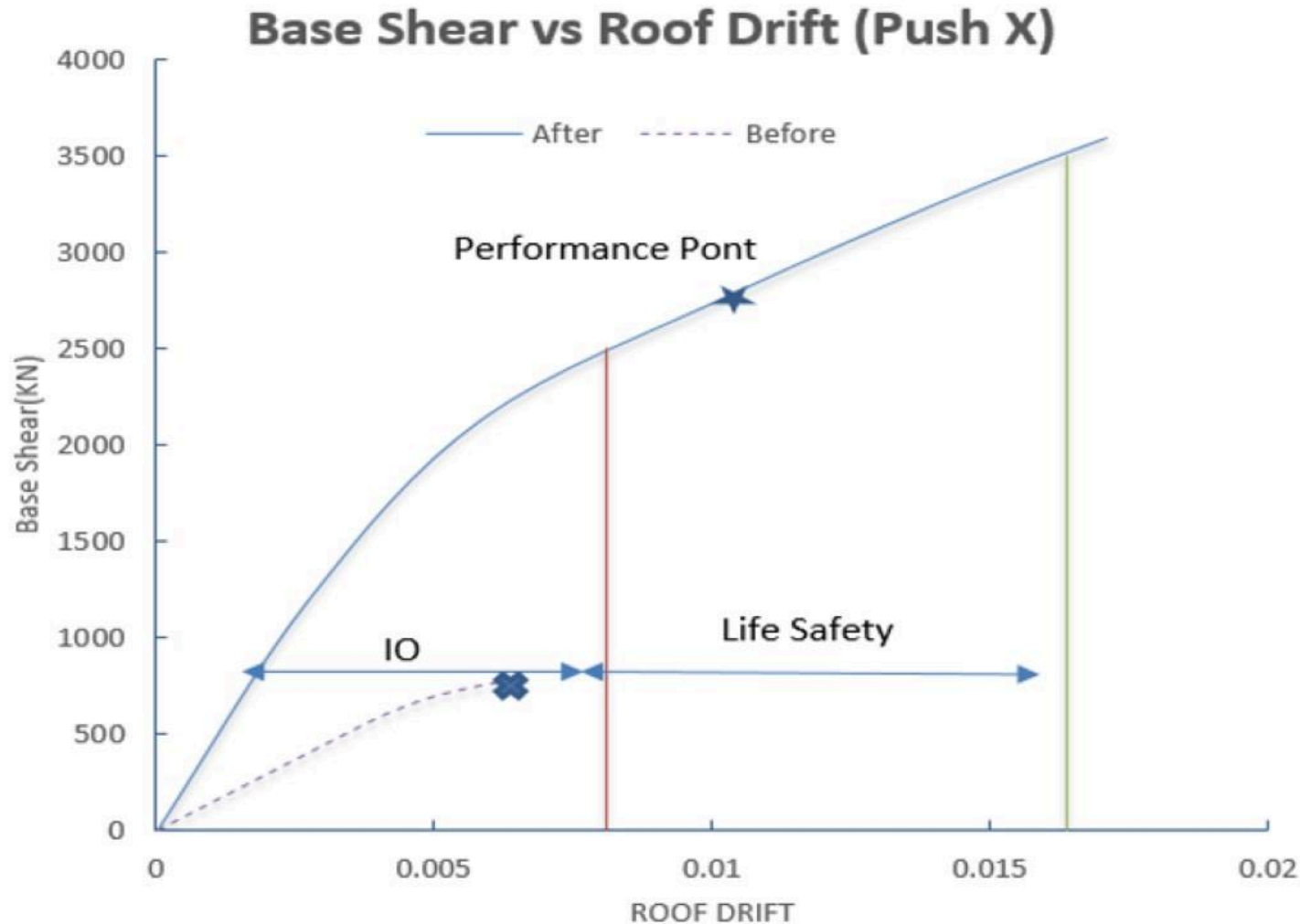


Existing Condition



After Retrofitting of Building

RC Buildings: Pushover Analysis



Comparing before and after Behavior



Retrofit Design

- **Setting Performance Objectives**
- **Selection of Appropriate Retrofitting Option**
- **Design of retrofitting elements**



Criteria for Retrofit Design

- **Minimum Life Safety at MMI IX Shaking**
- **Maximum 30% of New Construction Cost**

For all Options!!!



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Retrofitting Options for Masonry Buildings

Splint and Bandage Method

➤ Splint and Bandage in brick masonry wall



Splint and Bandage method

➤ Splint and Bandage in stone masonry wall



Jacketing method

➤ Concrete jacketing with steel bar mesh



Jacketing method

➤ Jacketing with G.I. wire mesh and cement plaster



Model Verification



PULL DOWN TEST



Non Retrofitted Building



Retrofitted Building



Retrofitted Buildings

- **NSET was involved in retrofitting of about 300 school buildings and some private residences before Gorkha earthquake**
- **All buildings survived with virtually no damage to the buildings**

Jacketing by PP Band Mesh



Polypropylene band (PP-band)



Why should PP-band be used?

- ◆ Cheap
- ◆ Worldwide available
- ◆ Strong and tolerates large deformations
- ◆ Durable
- ◆ Easy to handle and transport



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Model Verification





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Using steel sections



Using steel sections

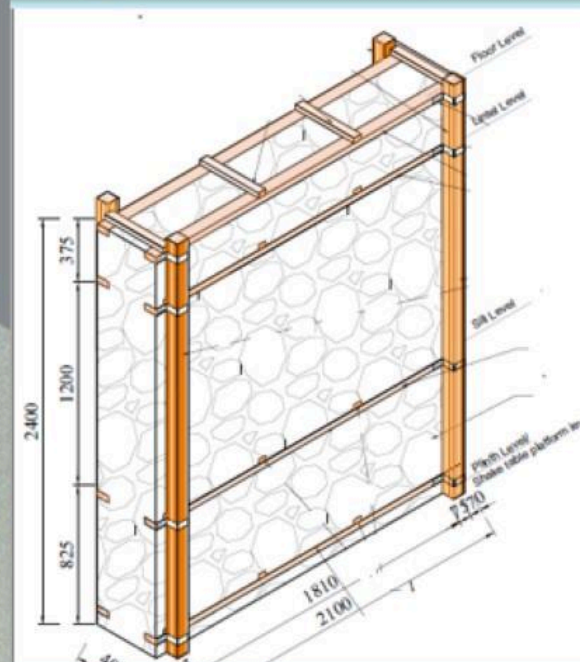
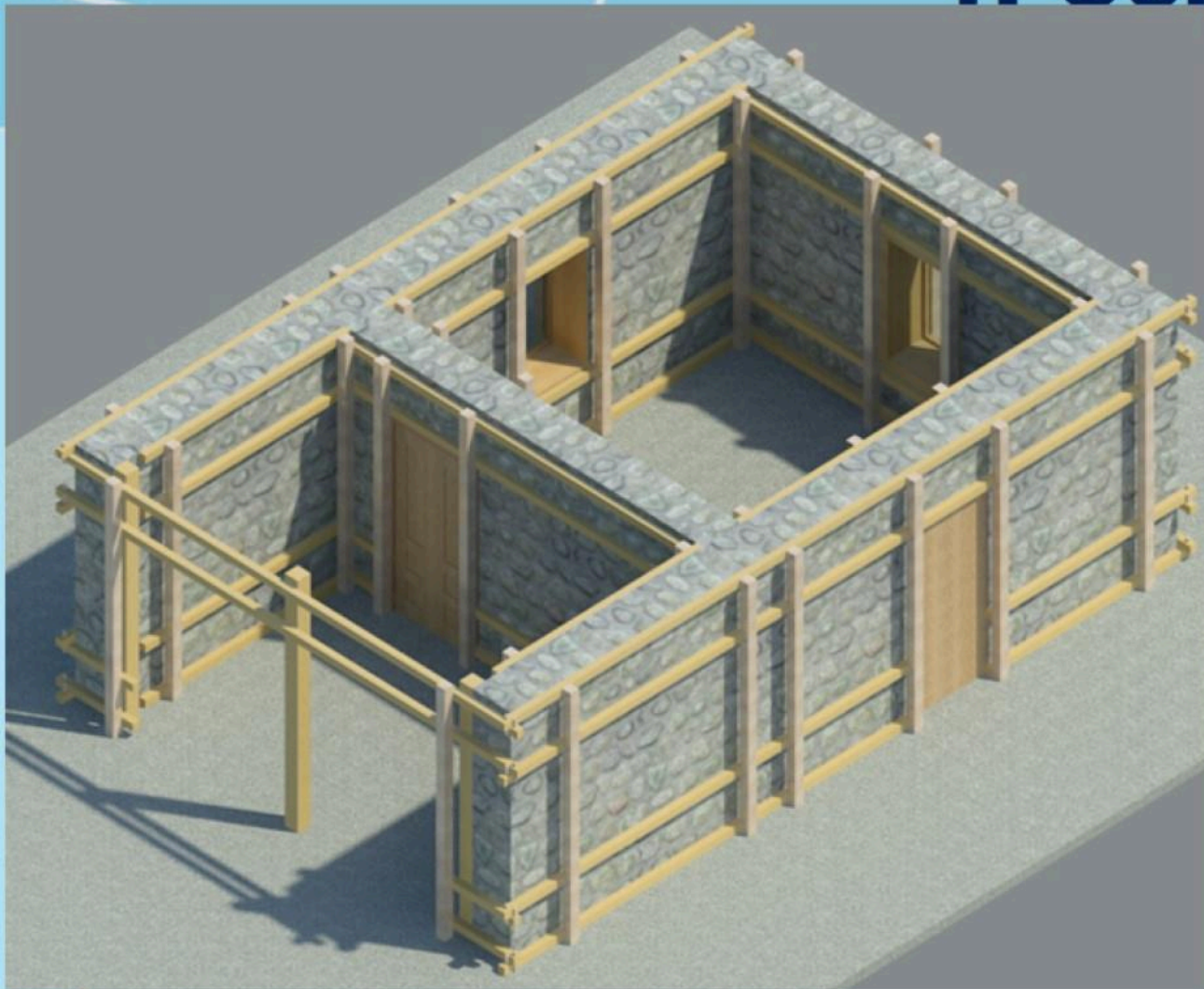




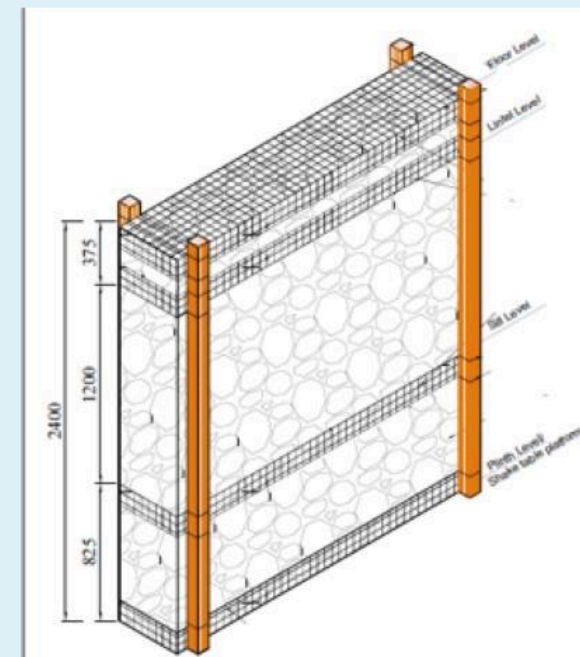
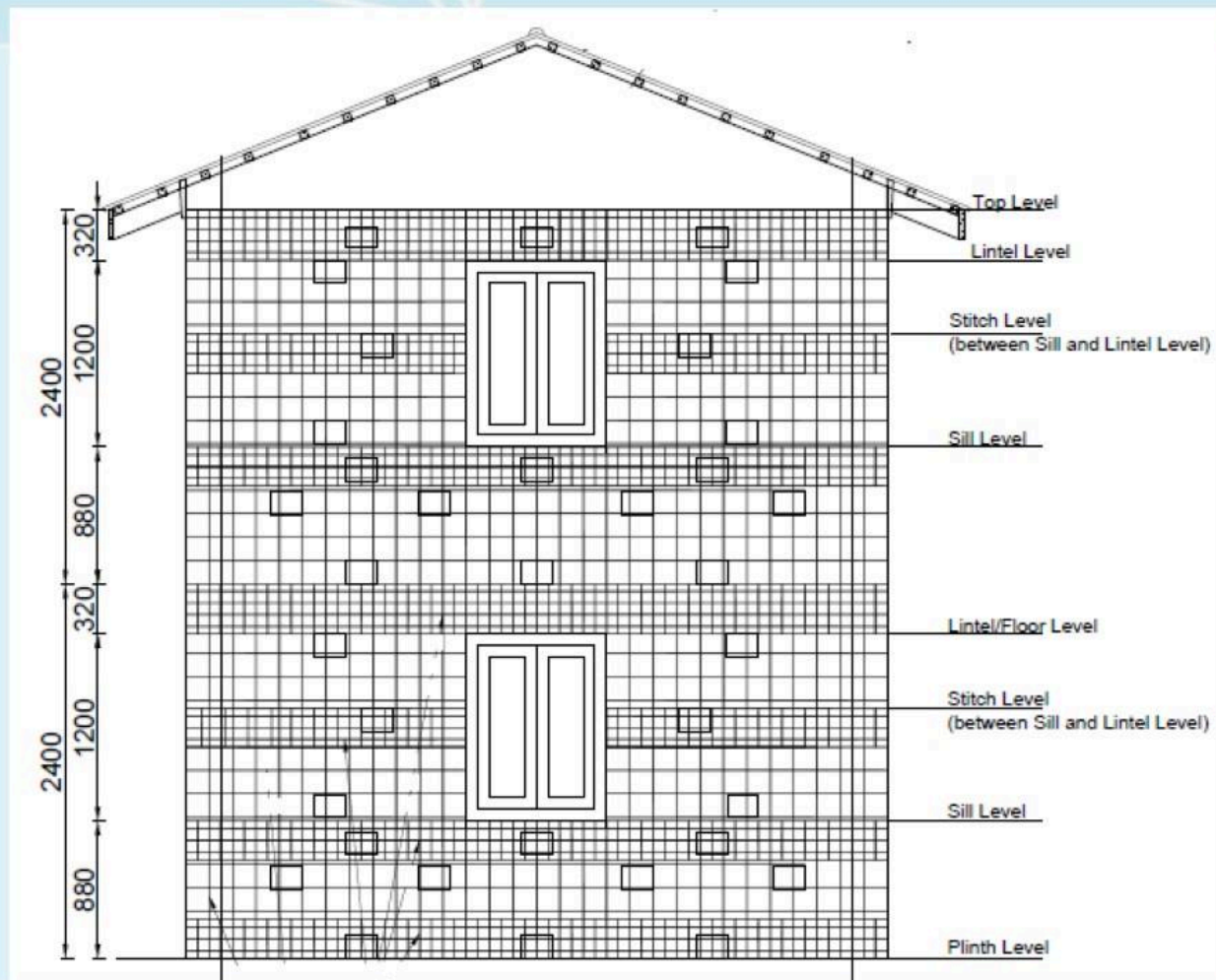
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Suggested Retrofitting Option for Low Strength Masonry

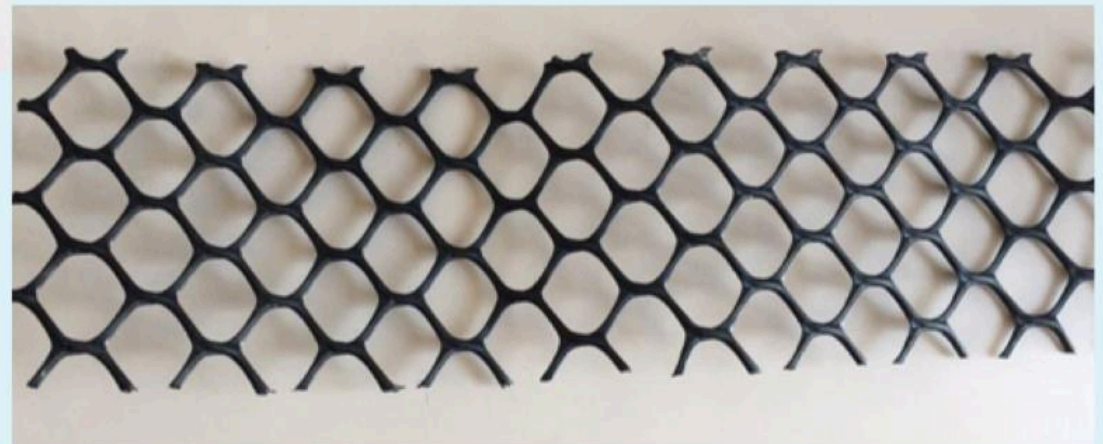
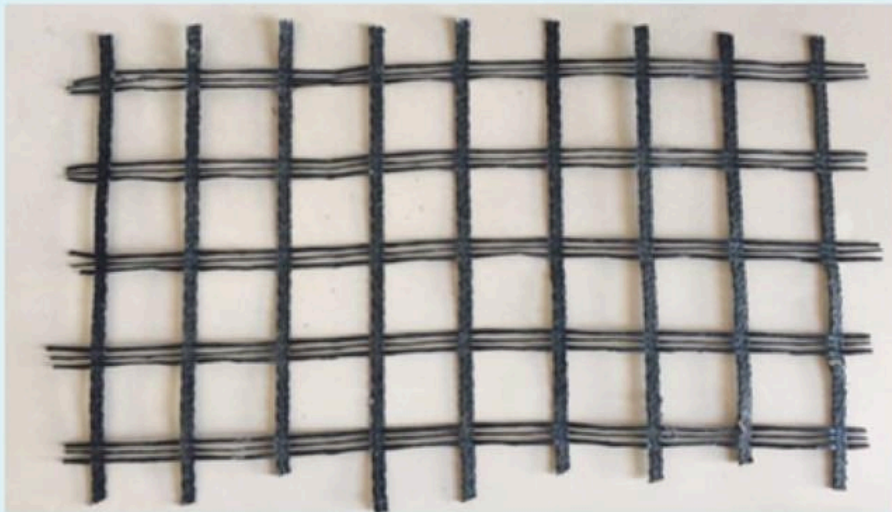
1. Using wooden bands



Using Gabion Wire Bandage



2. Using GeoMesh





Other common methods of retrofitting

- Improving plan configuration
- Elevation improvement
- Load path improvement
- Inserting new walls
- Modification of roofs or floors
- Reduction in building mass

Architectural Modification



Before

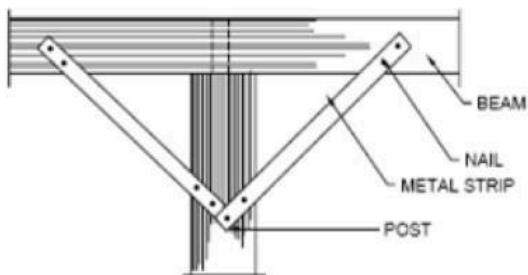
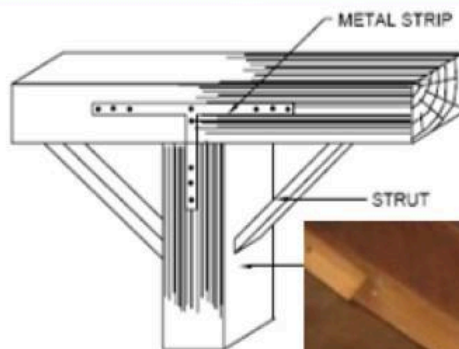
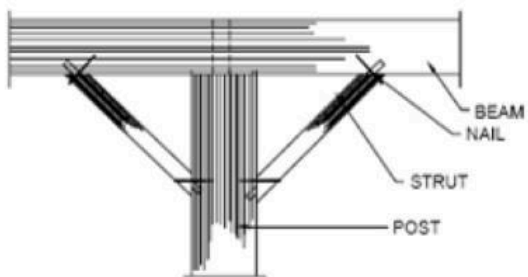


After



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Floor/Roof Bracing and Connection



Various Retrofitting Techniques RC Frame Buildings

Minimum thickness 100mm on all sides

Walls surrounding the columns need to be Demolished

Suitable for important buildings or need to add storey

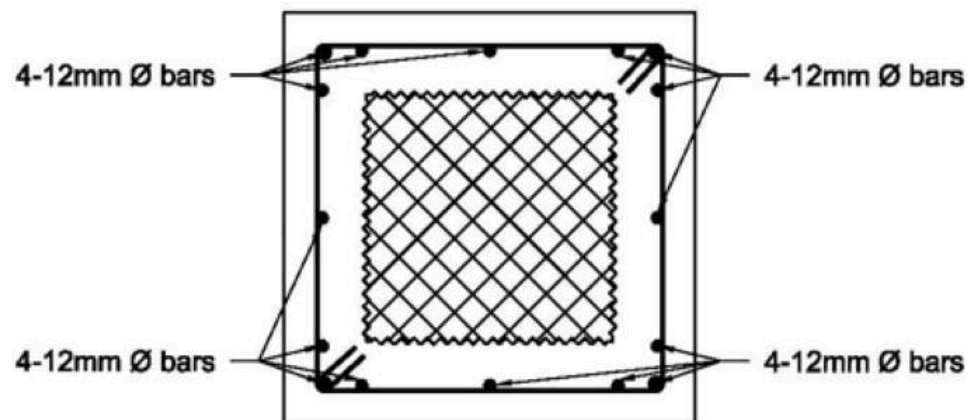
Generally beam strengthening is not required

Foundation strengthening is generally required

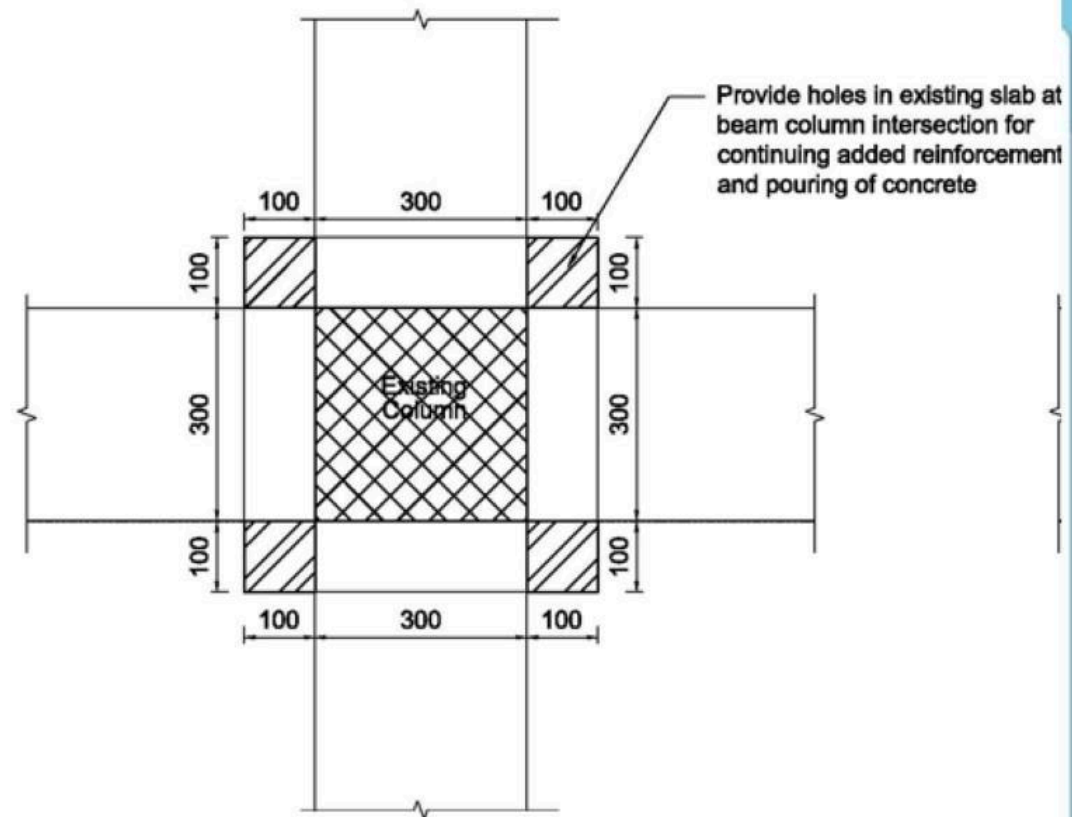
1. RC Jacketing of Columns



Jacketing of Columns



**REINFORCEMENT FOR
COLUMN JACKETING**



**DETAIL SHOWING PORTION OF
SLAB TO BE CUT**



RC Jacketing of Columns/ Beams

5 Storey

2. Steel Profile Jacketing



Does not increase stiffness but increases
Flexural capacity

Feasible only if column size is at least
300mm X300mm

Beam passing through the center of column

Only few required columns can be
strengthened

External Steel Caging...

3 Storey

3.Adding Wing Wall



The most suitable if the
thickness
In one side is 9'

Lateral stiffness and strength
can be
increased

Adequate connection between
column
And wall need to be ensured

Less intervention

Adding Wing Wall



Addition of Walls
adjacent to
Columns



Addition of
Reinforcing mesh



Micro-concreting on
the reinforced surface

2 Storey

4.Strengthening of Infill Masonry Walls



When columns are lean and non Ductile

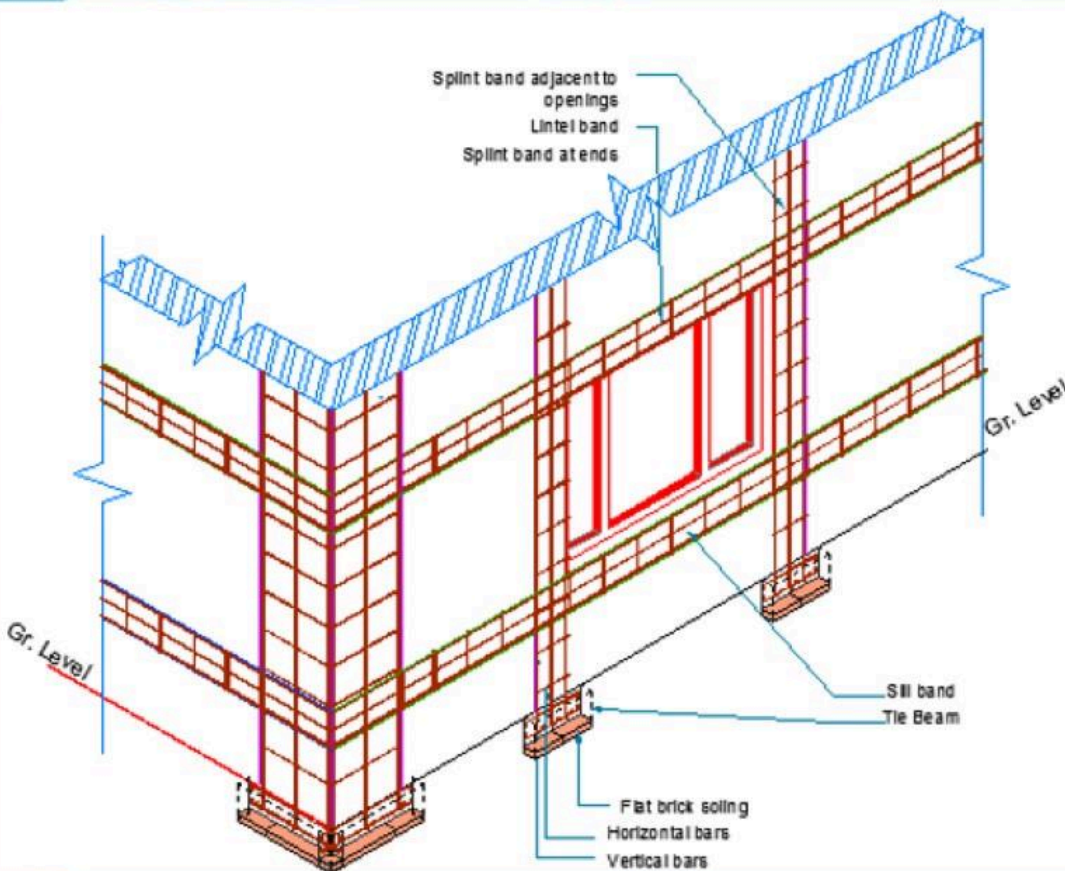
If infill walls are 9' thick contribution Of wall can be considered

Reinforcements are provided Only at critical locations in Splint And Bandage method

Wall jacketing is done if walls have Low strength

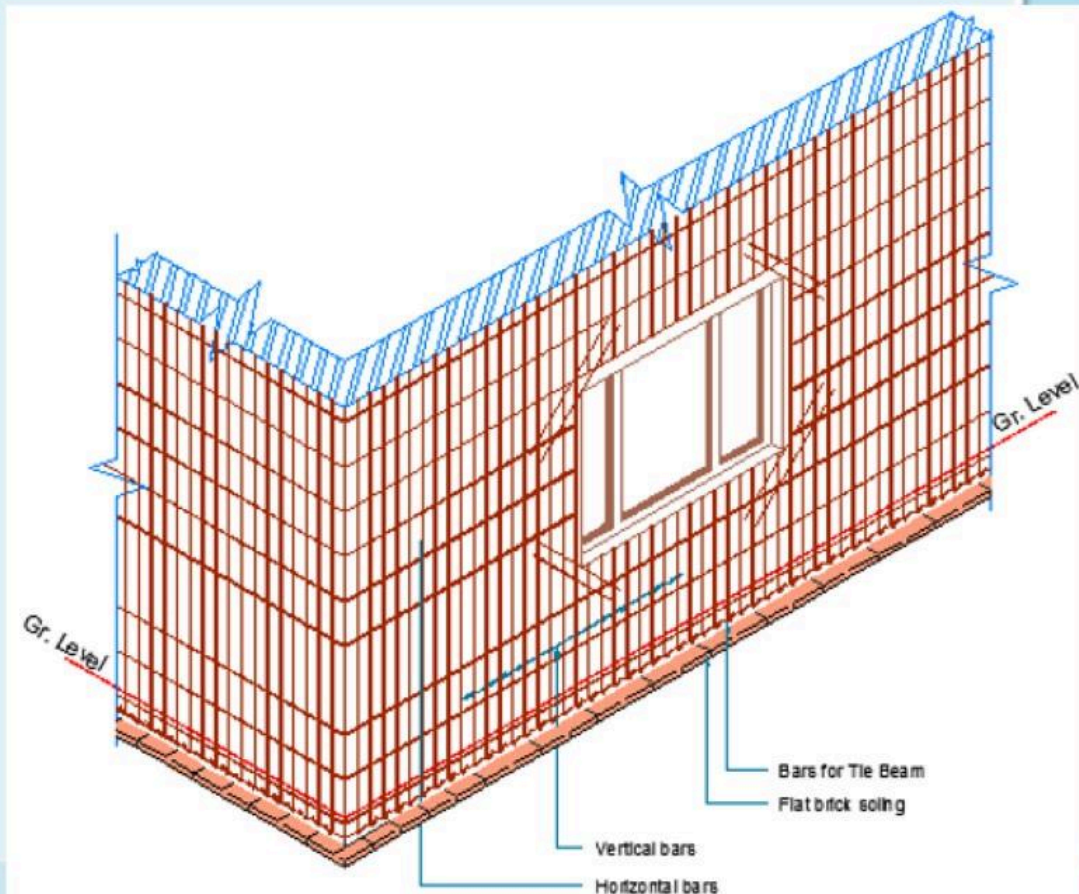
Splint and Bandage

Strengthening of Infill Masonry Walls



Splint and Bandage Technique

Jacketing of wall

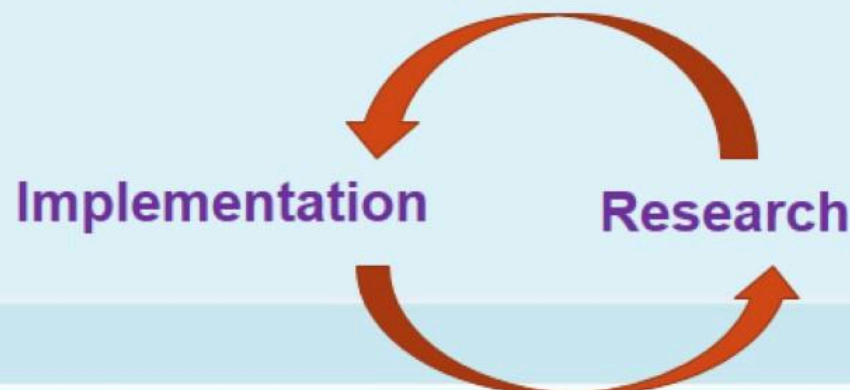


**TYPICAL VIEW OF JACKETING ON
OUTER WALL SURFACE**

SCALE = 1:40

Conclusions

- **Appropriate Seismic Retrofitting Option:**
 - **Technically Feasible**
 - **Economically Affordable (20-30% cost)**
 - **Local Available Materials can be used**
- **Replication: Slow than new construction**
- **More Economic/Appropriate Technology:
Continuous Implementation and Research**





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Thank You!